GLUING FINITELY MANY FINITE LOCAL EXTENSIONS ALONG THE HEIGHT ONE PRIMES OF A TWO-DIMENSIONAL LOCAL RING

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed. The author's label lemma-perpare-key contains a misspelling; it is reproduced verbatim in the claim key and in the source anchors and is not corrected.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The counted statement and proof contain no cross reference at all, so the assembled document defines every label it uses and no external reference stub is needed.

Allowed standard prerequisites (editorial).

- Noetherian local ring of dimension two with a nonzerodivisor t, its height one primes above t. Standard commutative algebra; the statement writes V(t) = {m, q_1, ..., q_r} for the primes containing t, so the q_i are the height one primes of the two-dimensional local ring. Supplies the geometry of the situation, in particular that only finitely many primes lie above t and that the localizations A_{q_i} are one-dimensional; no part of the gluing construction is taken from it.
- Localization is exact and A is Noetherian. Standard commutative algebra; used at proof lines 1122-1124 to conclude that M and M' are finite A-modules with M_{q_i} = M'_{q_i} = B_i. Supplies exactness of localization and finite generation; the construction of M, M' and B stays inside the counted proof.

Lemma 0.1. *Let $(A, \mathfrak{m})$ be a 2-dimensional Noetherian local ring. Let $t \in \mathfrak{m}$ be a nonzerodivisor. Say $V(t) = \{\mathfrak{m}, \mathfrak{q}_1, \ldots, \mathfrak{q}_r\}$. Let $A_{\mathfrak{q}_i} \subset B_i$ be a finite ring extension with $B_i/A_{\mathfrak{q}_i}$ annihilated by a power of t. Then there exists a finite extension $A \subset B$ of local rings identifying residue fields with $B_i \cong B_{\mathfrak{q}_i}$ and B/A annihilated by a power of t.*

Proof. Choose $n > 0$ such that $B_i \subset t^{-n}A_{\mathfrak{q}_i}$. Let $M \subset t^{-n}A$, resp. $M' \subset t^{-2n}A$ be the A-submodule consisting of elements mapping to B_i in $t^{-n}A_{\mathfrak{q}_i}$, resp. $t^{-2n}A_{\mathfrak{q}_i}$. Then $M \subset M'$ are finite A-modules as A is Noetherian and $M_{\mathfrak{q}_i} = M'_{\mathfrak{q}_i} = B_i$ as localization is exact. Thus M'/M is annihilated by $\mathfrak{m}^c$ for some $c > 0$. Observe that $M \cdot M \subset M'$ under the multiplication $t^{-n}A \times t^{-n}A \to t^{-2n}A$. Hence $B = A + \mathfrak{m}^{c+1}M$ is a finite A-algebra with the correct localizations. We omit the verification that B is local with maximal ideal $\mathfrak{m} + \mathfrak{m}^{c+1}M$. □

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